EA ELITE IAL MATHEMATICS

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 May/June 2022 Worked Solutions

MayJune 2022 paper rebuilt with the worked solution after each question.

10
paper questions

WMA11
Pure 1

Single-paper solutions
standalone IAL route

Dr Eslam Ahmed

Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 May/June 2022

MayJune 2022 | 10 questions | 75 marks

10
questions

75
marks

Answers
worked solution after each question

WMA11/01 MAY/JUNE 2022

Question 1

4 marks

Integration

1. Find

$$\int\left(10x^5 + 6x^3 - \frac{3}{x^2}\right)\mathrm{d}x$$

giving your answer in simplest form.

(4)

(Total 4 marks)

WMA11/01 MAY/JUNE 2022

Worked Solution - Question 1

Integration

Primary topic

1. Rewrite the negative power

$$\int\left(10x^5+6x^3-\frac{3}{x^2}\right)dx=\int(10x^5+6x^3-3x^{-2})dx.$$

2. Integrate term by term

$\int 10x^5dx=\dfrac{10x^6}{6}=\dfrac{5}{3}x^6$, $\int 6x^3dx=\dfrac{6x^4}{4}=\dfrac{3}{2}x^4$, and $\int -3x^{-2}dx=3x^{-1}$.

3. Add the constant

Therefore the integral is $\dfrac{5}{3}x^6+\dfrac{3}{2}x^4+\dfrac{3}{x}+c$.

Final answer

$\dfrac{5}{3}x^6+\dfrac{3}{2}x^4+\dfrac{3}{x}+c$.

Question 2

5 marks

Basic Trigonometry

2. In the triangle ABC,

- $AB = 21\text{ cm}$
- $BC = 13\text{ cm}$
- angle $BAC = 25°$
- angle $ACB = x°$

(a) Use the sine rule to find the value of $\sin x°$, giving your answer to 4 decimal places.
(2)

Given also that AB is the longest side of the triangle,

(b) find the value of x, giving your answer to 2 decimal places.
(3)

(Total 5 marks)

Worked Solution - Question 2

Basic Trigonometry

Primary topic

1. Use the sine rule

The side opposite x is $AB = 21$, and the side opposite $25°$ is $BC = 13$, so $\dfrac{\sin x}{21} = \dfrac{\sin 25°}{13}$.

2. Find the sine value

$$\sin x = \frac{21\sin 25°}{13} = 0.68269\ldots.$$

3. Choose the correct angle

The angle from the calculator is $43.05°$, but AB is the longest side, so x is obtuse.

Hence $x = 180° - 43.05° = 136.95°$.

Final answer

$x = 137°$ to 3 significant figures.

WMA11/01 MAY/JUNE 2022

Question 3

5 marks

Indices & Surds

3. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(i) Show that $\dfrac{\sqrt{180}-\sqrt{80}}{\sqrt{5}}$ is an integer and find its value.

(2)

(ii) Simplify

$$\frac{4\sqrt{5}-5}{7-3\sqrt{5}}$$

giving your answer in the form $a+b\sqrt{5}$ where a and b are rational numbers.

(3)

(Total 5 marks)

Worked Solution - Question 3

Indices & Surds

Primary topic

1. Simplify the square roots

$\sqrt{180} = 6\sqrt{5}$ and $\sqrt{80} = 4\sqrt{5}$.

2. Complete part (a)

$$\frac{\sqrt{180} - \sqrt{80}}{\sqrt{5}} = \frac{6\sqrt{5} - 4\sqrt{5}}{\sqrt{5}} = 2.$$

3. Rationalise part (b)

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}} \times \frac{7 + 3\sqrt{5}}{7 + 3\sqrt{5}} = \frac{(4\sqrt{5} - 5)(7 + 3\sqrt{5})}{49 - 45}.$$

4. Expand and simplify

The numerator is $28\sqrt{5} + 60 - 35 - 15\sqrt{5} = 25 + 13\sqrt{5}$, so the expression is $\dfrac{25 + 13\sqrt{5}}{4}$.

Final answer

$(a)\ 2.$

$(b)\ \dfrac{25}{4} + \dfrac{13}{4}\sqrt{5}.$

Question 4

6 marks

Transformations

4.

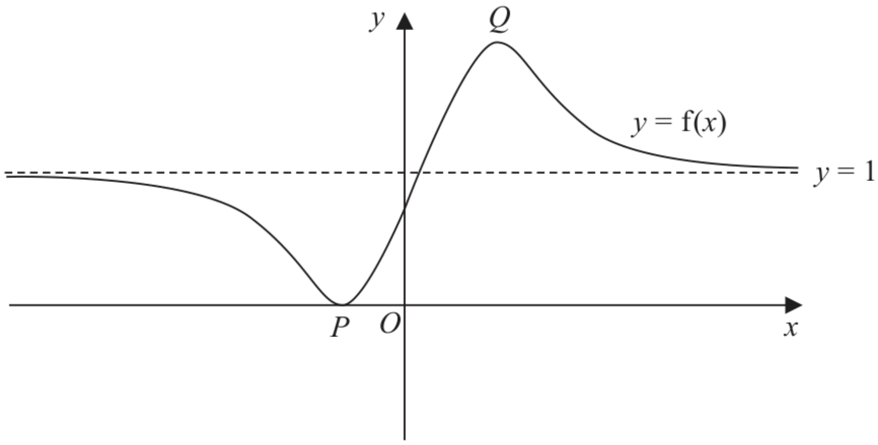

Figure 1

Figure 1 shows a sketch of a curve with equation $y = \mathrm{f}(x)$

The curve has a minimum at $P(-1, 0)$ and a maximum at $Q\left(\frac{3}{2}, 2\right)$

The line with equation $y = 1$ is the only asymptote to the curve.

On separate diagrams sketch the curves with equation

(i) $y = \mathrm{f}(x) - 2$

(3)

(ii) $y = \mathrm{f}(-x)$

(3)

On each sketch you must clearly state

- the coordinates of the maximum and minimum points
- the equation of the asymptote

(Total 6 marks)

Worked Solution - Question 4

Transformations

Primary topic

1. Translate for y = f(x) - 2

Subtracting 2 moves every point of the graph down by 2 units.

2. State the new key features

The minimum $P(-1, 0)$ becomes $(-1, -2)$, the maximum $Q(\frac{3}{2}, 2)$ becomes $(\frac{3}{2}, 0)$, and the asymptote $y = 1$ becomes $y = -1$.

3. Reflect for y = f(-x)

Replacing x by $-x$ reflects the graph in the y-axis.

4. State the reflected key features

The minimum $P(-1, 0)$ becomes $(1, 0)$, the maximum $Q(\frac{3}{2}, 2)$ becomes $(-\frac{3}{2}, 2)$, and the horizontal asymptote remains $y = 1$.

Final answer

(i) minimum $(-1, -2)$, maximum $(\frac{3}{2}, 0)$, asymptote $y = -1$. (ii) minimum $(1, 0)$, maximum $(-\frac{3}{2}, 2)$, asymptote $y = 1$.

WMA11/01 MAY/JUNE 2022

Question 5

9 marks

Straight Line

5. The curve C has equation $y = \mathrm{f}(x)$

Given that

- $\mathrm{f}(x)$ is a quadratic expression
- the maximum turning point on C has coordinates $(-2, 12)$
- C cuts the negative x-axis at -5

(a) find $\mathrm{f}(x)$

(4)

The line l_1 has equation $y = \dfrac{4}{5}x$

Given that the line l_2 is perpendicular to l_1 and passes through $(-5, 0)$

(b) find an equation for l_2, writing your answer in the form $y = mx + c$ where m and c are constants to be found.

(3)

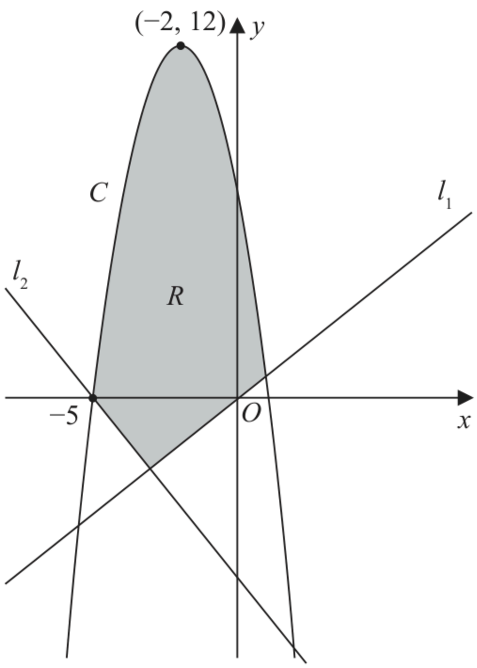

Figure 2

Figure 2 shows a sketch of the curve C and the lines l_1 and l_2

(c) Define the region R, shown shaded in Figure 2, using inequalities.

(2)

(Total 9 marks)

WMA11/01 MAY/JUNE 2022

Worked Solution - Question 5

Straight Line

Primary topic

1. Use symmetry of the quadratic

The maximum is at $x = -2$, and one root is $x = -5$. The other root is the same distance to the right of -2, so it is $x = 1$.

2. Find the quadratic

Let $f(x) = a(x+5)(x-1)$. Since the maximum point is $(-2, 12)$, $12 = a(3)(-3) = -9a$, so $a = -\dfrac{4}{3}$.

3. Find the perpendicular gradient

The gradient of l_1 is $\dfrac{4}{5}$, so the gradient of a perpendicular line is $-\dfrac{5}{4}$.

4. Find l2

Using the point $(-5, 0)$, $y = -\dfrac{5}{4}x + c$. Then $0 = \dfrac{25}{4} + c$, so $c = -\dfrac{25}{4}$ and l_2 is $y = -\dfrac{5}{4}x - \dfrac{25}{4}$.

5. Write the inequalities for R

The region is below the curve and above both straight lines, so $y \leq f(x)$, $y \geq \dfrac{4}{5}x$, and $y \geq -\dfrac{5}{4}x - \dfrac{25}{4}$.

Final answer

$(a)\ f(x) = -\dfrac{4}{3}(x+5)(x-1).$

$(b)\ y = -\dfrac{5}{4}x - \dfrac{25}{4}.$

$(c)\ y \leq f(x),\ y \geq \dfrac{4}{5}x,\ y \geq -\dfrac{5}{4}x - \dfrac{25}{4}.$

Question 6

7 marks

Quadratics

6. **In this question you must show all stages of your working.**

Solutions relying on calculator technology are not acceptable.

(a) Given that

$$2xy - 3x^2 = 50$$

and

$$y - x^3 + 6x = 0$$

show that

$$2x^4 - 15x^2 - 50 = 0$$

(2)

(b) Hence solve the simultaneous equations

$$2xy - 3x^2 = 50$$
$$y - x^3 + 6x = 0$$

Give your answers in fully simplified surd form.

(5)

(Total 7 marks)

WMA11/01 MAY/JUNE 2022

Worked Solution - Question 6

Quadratics

Primary topic

1. Rearrange the second equation

From $y - x^3 + 6x = 0$, we get $y = x^3 - 6x$.

2. Substitute into the first equation

$2x(x^3 - 6x) - 3x^2 = 50$, so $2x^4 - 15x^2 - 50 = 0$.

3. Solve the quadratic in x squared

Let $u = x^2$. Then $2u^2 - 15u - 50 = 0$, giving $u = 10$ or $u = -\frac{5}{2}$.

4. Reject the negative value

Since x^2 cannot be negative, $x^2 = 10$, so $x = \pm\sqrt{10}$.

5. Find y

$y = x^3 - 6x = x(x^2 - 6) = x(10 - 6) = 4x$. Therefore $y = 4\sqrt{10}$ when $x = \sqrt{10}$ and $y = -4\sqrt{10}$ when $x = -\sqrt{10}$.

Final answer

$(x, y) = (\sqrt{10}, 4\sqrt{10})$ or $(-\sqrt{10}, -4\sqrt{10})$.

Question 7

9 marks

Integration

7. The curve C has equation $y = \mathrm{f}(x),\ x > 0$

Given that

- $\mathrm{f}'(x) = \dfrac{2}{\sqrt{x}} + \dfrac{A}{x^2} + 3$, where A is a constant
- $\mathrm{f}''(x) = 0$ when $x = 4$

(a) find the value of A.

(4)

Given also that

- $\mathrm{f}(x) = 8\sqrt{3}$, when $x = 12$

(b) find $\mathrm{f}(x)$, giving each term in simplest form.

(5)

(Total 9 marks)

WMA11/01 MAY/JUNE 2022

Worked Solution - Question 7

Integration

Primary topic

1. Differentiate f'(x)

$f'(x) = 2x^{-1/2} + Ax^{-2} + 3$, so $f''(x) = -x^{-3/2} - 2Ax^{-3}$.

2. Use f''(4) = 0

$-4^{-3/2} - 2A(4^{-3}) = 0$, so $-\dfrac{1}{8} - \dfrac{A}{32} = 0$ and $A = -4$.

3. Substitute A into f'(x)

$f'(x) = \dfrac{2}{\sqrt{x}} - \dfrac{4}{x^2} + 3$.

4. Integrate to find f(x)

$f(x) = 4\sqrt{x} + \dfrac{4}{x} + 3x + c$.

5. Use f(12) = 8sqrt3

$f(12) = 8\sqrt{3} + \dfrac{1}{3} + 36 + c = 8\sqrt{3}$, so $c = -\dfrac{109}{3}$.

Final answer

$f(x) = 4\sqrt{x} + \dfrac{4}{x} + 3x - \dfrac{109}{3}$.

WMA11/01 MAY/JUNE 2022

Question 8

10 marks

Radians

8.

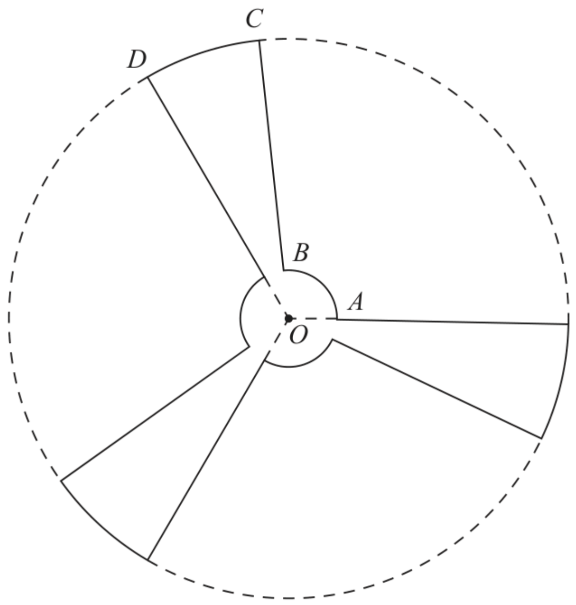

Figure 3

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

The fan consists of three identical sections congruent to *OABCDO*, shown in Figure 3, where

- *OABO* is a sector of a circle with centre *O* and radius 9 cm
- *OBCDO* is a sector of a circle with centre *O* and radius 84 cm
- angle $AOD = \dfrac{2\pi}{3}$ radians

Given that the length of the arc *AB* is 15 cm,

(a) show that the length of the arc *CD* is 35.9 cm to one decimal place.

(3)

The face of the fan is modelled to be a flat surface.

Find, according to the model,

(b) the perimeter of the face of the fan, giving your answer to the nearest cm,

(2)

(c) the surface area of the face of the fan.

Give your answer to 3 significant figures and make your units clear.

(5)

(Total 10 marks)

WMA11/01 MAY/JUNE 2022

Worked Solution - Question 8

Radians

Primary topic

1. Find the small sector angle

For arc AB, $s = r\theta$, so $15 = 9\theta$ and $\theta = \dfrac{5}{3}$ radians.

2. Find the outer arc angle

Each identical section has total angle $\dfrac{2\pi}{3}$, so the angle for arc CD is $\dfrac{2\pi}{3} - \dfrac{5}{3} = \dfrac{2\pi - 5}{3}$.

3. Find CD

$CD = 84\left(\dfrac{2\pi - 5}{3}\right) = 28(2\pi - 5) = 35.9\ \mathrm{cm}$ to 3 significant figures.

4. Find the perimeter

There are 3 inner arcs, 3 outer arcs, and 6 straight radial edges of length $84 - 9 = 75$. Thus $P = 3(15) + 3(35.929\ldots) + 6(75) = 602.8\ldots\ \mathrm{cm}$.

5. Find the area of one section

One section has area $\dfrac{1}{2}(9^2)\left(\dfrac{5}{3}\right) + \dfrac{1}{2}(84^2)\left(\dfrac{2\pi - 5}{3}\right) = 1576.525\ldots\ \mathrm{cm}^2$.

6. Find the total area

The total area is $3(1576.525\ldots) = 4729.57\ldots\ \mathrm{cm}^2 = 4.73 \times 10^3\ \mathrm{cm}^2$ to 3 significant figures.

Final answer

(a) $CD = 35.9$ cm.

(b) 603 cm.

(c) $4.73 \times 10^3 \text{ cm}^2$.

WMA11/01 MAY/JUNE 2022

Question 9

8 marks

Trigonometric Functions

9.

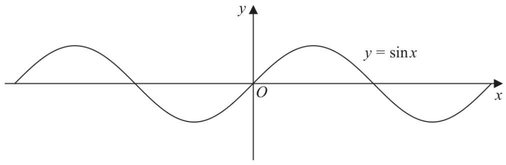

Figure 4

Figure 4 shows part of the graph of the curve with equation $y = \sin x$

Given that $\sin\alpha = p$, where $0 < \alpha < 90°$

(a) state, in terms of p, the value of

(i) $2\sin(180° - \alpha)$

(ii) $\sin(\alpha - 180°)$

(iii) $3 + \sin(180° + \alpha)$

(3)

A copy of Figure 4, labelled Diagram 1, is shown on page 27.

On Diagram 1,

(b) sketch the graph of $y = \sin 2x$

(2)

(c) Hence find, in terms of α, the x coordinates of any points in the interval $0 < x < 180°$ where

$$\sin 2x = p$$

(3)

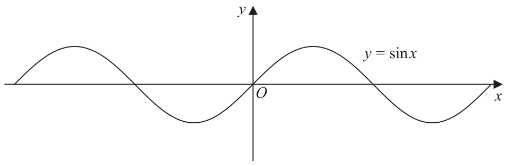

Diagram 1

(Total 8 marks)

WMA11/01 MAY/JUNE 2022

Worked Solution - Question 9

Trigonometric Functions

Primary topic

1. Use sine symmetry

$\sin(180° - \alpha) = \sin\alpha = p$, so $2\sin(180° - \alpha) = 2p$.

2. Use odd symmetry

$\sin(\alpha - 180°) = -\sin(180° - \alpha) = -p$.

3. Use the third quadrant identity

$\sin(180° + \alpha) = -\sin\alpha = -p$, so $3 + \sin(180° + \alpha) = 3 - p$.

4. Sketch y = sin 2x

The graph has amplitude 1 and period $180°$. It crosses the x-axis at $0°, 90°, 180°$, has a maximum at $45°$, and a minimum at $135°$.

5. Solve sin 2x = p

Since $\sin\alpha = p$ and $0 < 2x < 360°$, the possible values are $2x = \alpha$ or $2x = 180° - \alpha$.

6. Find x

Therefore $x = \dfrac{\alpha}{2}$ or $x = 90° - \dfrac{\alpha}{2}$.

Final answer

$(a)(i)\ 2p,\ (ii)\ -p,\ (iii)\ 3-p.$

$(c)\ x = \dfrac{\alpha}{2},\ 90° - \dfrac{\alpha}{2}.$

WMA11/01 MAY/JUNE 2022

Question 10

12 marks

Differentiation

10.

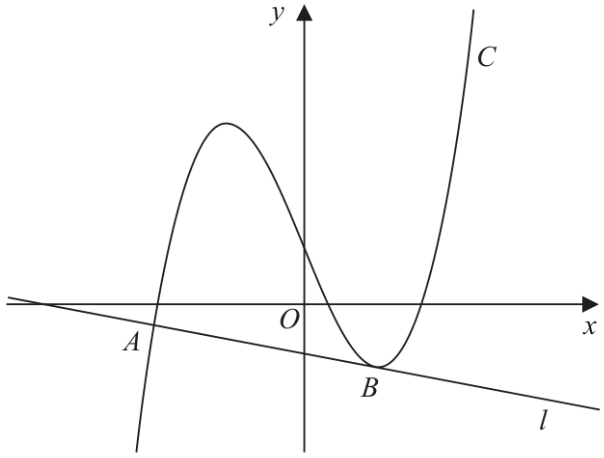

Figure 5

Figure 5 shows a sketch of the curve C with equation

$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where k is a constant.

(a) Find $\dfrac{dy}{dx}$

(2)

The line l, shown in Figure 5, is the normal to C at the point A with x coordinate $-\dfrac{7}{2}$

Given that l is also a tangent to C at the point B,

(b) show that the x coordinate of the point B is a solution of the equation

$$12x^2 + 4x - 33 = 0$$

(4)

(c) Hence find the x coordinate of B, justifying your answer.

(2)

Given that the y intercept of l is -1

(d) find the value of k.

(4)

(Total 12 marks)

WMA11/01 MAY/JUNE 2022

Worked Solution - Question 10

Differentiation

Primary topic

1. Differentiate the curve

For $y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$, $\frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}$.

2. Find the tangent gradient at A

At $x = -\frac{7}{2}$, $\frac{dy}{dx} = \frac{6}{7}\left(\frac{49}{4}\right) + \frac{2}{7}\left(-\frac{7}{2}\right) - \frac{5}{2} = \frac{21}{2} - 1 - \frac{5}{2} = 7$.

3. Find the gradient of l

Since l is the normal at A, its gradient is the negative reciprocal of 7, so $m_l = -\frac{1}{7}$.

4. Use that l is tangent at B

At B, the tangent gradient is also $-\frac{1}{7}$, so $\frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2} = -\frac{1}{7}$.

5. Reach the given quadratic

Multiplying by 14 gives $12x^2 + 4x - 35 = -2$, hence $12x^2 + 4x - 33 = 0$.

6. Solve for the x-coordinate of B

$12x^2 + 4x - 33 = 0$ gives $x = \frac{-4 \pm 40}{24}$, so $x = \frac{3}{2}$ or $x = -\frac{11}{6}$. From the diagram, B is the point to the right of the y-axis, so $x_B = \frac{3}{2}$.

7. Use the y-intercept of l

Since the y-intercept is -1 and the gradient is $-\frac{1}{7}$, the line is $y = -\frac{1}{7}x - 1$.

8. Find the y-coordinate of A

At $x = -\frac{7}{2}$ on the line, $y = -\frac{1}{7}\left(-\frac{7}{2}\right) - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$.

9. Substitute A into the curve

$$-\frac{1}{2}=\frac{2}{7}\left(-\frac{7}{2}\right)^3+\frac{1}{7}\left(-\frac{7}{2}\right)^2-\frac{5}{2}\left(-\frac{7}{2}\right)+k=-\frac{7}{4}+k.$$

10. Find k

Therefore $k=\frac{5}{4}$.

Final answer

$(a)\ \frac{dy}{dx}=\frac{6}{7}x^2+\frac{2}{7}x-\frac{5}{2}.$

$(c)\ x_B=\frac{3}{2}.$

$(d)\ k=\frac{5}{4}.$